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RESEARCH ARTICLE



Screening, identification and evaluation of *Trichoderma* spp. for biocontrol potential of common bean damping-off pathogens

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ABSTRACT

Damping-off caused by soil-borne pathogenic fungi is the major constraint responsible for significantly reduced yields and severe economic losses during common bean cultivation. In a search for alternative control measures, this study describes the investigation of 85 *Trichoderma* isolates obtained from bean rhizospheric soil to identify an environment friendly method of biological control of damping-off. The *Trichoderma* isolates were screened for their antagonistic potential against four common bean pathogens namely *Fusarium oxysporum*, *Fusarium solani*, *Macrophomina phaseolina* and *Pythium ultimum*. As a result, 6 *Trichoderma* isolates were selected and further identified as *T. asperellum* It-13, *T. koningiopsis* It-21, *T. erinaceum* It-58, *T. gamsii* It-62, *T. afroharzianum* P-8 and *T. harzianum* P-11 by amplifying the sequences of ITS1 and ITS4. Identified strains were found to produce volatile and non-volatile inhibitory metabolites as well as hydrolytic enzymes (chitinase, cellulase, protease and lipase) with *T. gamsii* It-62 exhibiting globally the highest enzyme secretion. Seeds treatment with *T. gamsii* showed a beneficial effect on the emergence of bean plants by inhibiting damping-off incidence and delaying the onset of disease symptoms induced by the different plant pathogens. As far as we know, this is the first report in Cameroon referring to the antagonistic effect of *Trichoderma gamsii* on soil-borne pathogens complex causing damping-off of common bean. It could be suggested that *T. gamsii* It-62 should be further investigated as a promising alternative to control common bean damping-off caused by plant pathogens.

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1. Introduction

Common bean (*Phaseolus vulgaris*) is one of the most important food legume for direct human consumption grown worldwide (Buruchara, 2006). In subtropical countries

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including Cameroon, beans are the primary source of dietary protein, and are widely referred to as the 'meat of the poor' (CIAT, 2013). Beyond promoting food, health and nutritional security, the value of the crop sales exceeds US\$ 500 million annually (CIAT, 2013; FAO, 2011).

Common bean is affected by several soil-borne plant pathogens resulting in yield losses of up to 80% (Park & Tu, 1994). Bean damping-off is one of the most important diseases caused by a complex of soil-borne pathogens which include species of *Fusarium*, *Pythium*, *Macrophomina* and *Rhizoctonia* (Hagerty et al., 2015; Mwang'ombe et al., 2008; Nzungize, Lyumugabe, Busogoro, & Baudoin, 2012; Schneider, Grafton, & Kelly, 2001). These pathogens attack beans at all growth stages, causing damping-off at the seedling stage, yellowing of the leaves and stunted growth (Nzungize et al., 2012).

Amongst strategies used to control damping-off, chemical practice is normally recommended. However, the use of broad-spectrum chemical pesticides on farms likely leads to increased environmental contamination, risks to humans and livestock and increases production costs (Mboussi, Musongong, & Nounamo, 2012). In order to manage plant pathogens by environment-friendly approaches, the use of rhizospheric microorganisms as biological tools is an important alternative (Pradhan & Sukla, 2005). *Trichoderma* spp. have been widely studied (Belete, Ayalew, & Ahmed, 2015; Harman, Howell, Viterbo, Chet, & Lorito, 2004; Hermosa, Viterbo, Chet, & Monte, 2012) and the genus *Trichoderma* are filamentous fungi reported to be widely distributed in various ecosystems. This ubiquitous genus has several species characterised by a high competitive nature for space and nutrient, reproductive ability and capacity to promote plant growth. *Trichoderma* species are effectively used as antagonistic agents to control several pathogens in many important crops (Abo-Elyousr, Sobhy, Abdel-Hafez, & Abdel-Rahim, 2014; Carvalho, Junior, Martins, Inglis, & Mello, 2014; Tchameni et al., 2017; Zhang et al., 2013). Most *Trichoderma* strains are more competent in controlling some pathogens, though others may be largely ineffective against some fungi (Hermosa et al., 2012). Therefore, screening of biocontrol strain is essential for their further development and commercialisation (Chen, Chen, & Chou, 2018).

The present study was carried out to screen, identify and evaluate potential strain of *Trichoderma* spp. for biocontrol *Fusarium oxysporum*, *Fusarium solani*, *Macrophomina phaseolina* and *Pythium ultimum* causing common bean damping-off.

2. Materials and methods

2.1. Microbial isolates

Rhizospheric soil samples were collected from bean farm at Soa (Center Region, Cameroon) during August to September 2014. Isolation of *Trichoderma* was made using serial dilution technique. Each collected soil sample was thoroughly mixed, grounded using of mortar and pestle, and passed through a 0.5 mm diameter soil screen sieve. One gram of soil sample was suspended in 9 ml sterile distilled water, homogenised by agitation and further serial dilutions of 10^{-2} , 10^{-3} and 10^{-4} were prepared. One millilitre of diluted suspension was introduced into potato dextrose agar medium in Petri plates and completely mixed by carefully swirling to uniformly spread the suspension. Emerging mycelia of fungal colonies were replicated after 48 h of incubation into new PDA medium and purified by successive transfers.

F. oxysporum, *F. solani*, *M. phaseolina* and *P. ultimum* used in this study were obtained from the collection of the Laboratory for Phytobiochemistry and Medicinal Plants Studies (Antimicrobial and Biocontrol Agents Unit) at University of Yaoundé I (Cameroon). These pathogens were previously isolated from root tissues of infected bean plants showing the typical symptoms of damping-off disease and the pathogenicity was estimated (Haggag & El-Gamal, 2012).

2.2. Screening for antagonistic activity of *Trichoderma* isolates

The dual culture method was performed *in vitro* to evaluate interactions between the antagonist and pathogen. Five mm diameter mycelial discs of a 5-day-old *Trichoderma* and pathogen cultures were placed on opposite sides and at equal distance from the periphery of the same PDA plate. Control plates were made by placing a sterile agar disc at opposite side of the pathogen inoculated disc. Each treatment consisted of three replicates and the experiment repeated twice. After 6 days of growth at 28°C with photoperiod of 12 h, the plates were examined for development of pathogens and growth inhibition percentage (%I) was determined according to the following equation:

$$\%I = \left(\frac{D_o - D}{D_o} \right) \times 100, \quad (1)$$

where D is the colony radial growth (mm) of pathogen in the presence of *Trichoderma* and D_o the radial growth (mm) in control.

The level of antagonism of each *Trichoderma* isolate in dual culture was recorded using the rating scale of R1 to R5 as described by Bell, Wells, and Markham (1982). R1 being the rate when *Trichoderma* completely invaded the pathogen colony and covered all the entire PDA plate; R2 the rate when it colonises at least two-thirds of the medium surface. For R3, the pathogen and antagonist each colonised approximately one-half of the medium. Concerning R4, the pathogens overgrew *Trichoderma* and covered the two-thirds of the culture medium while R5 it colonised all the entire plate.

2.3. Molecular identification of *Trichoderma* isolates

Trichoderma was first grown for 5 days on PDA plate at 25°C and the mycelia mat scraped from the surface of culture. DNA was extracted by adding 10 mg of mycelia in 100 µL lysis buffer (50 mM Tris-HCl pH 7.5, 3% SDS, 50 mM EDTA, 1% 2-mercaptoethanol) followed by microwave miniprep procedure according to Goodwin and Lee (1993). Thereafter, the final DNA pellet obtained was suspended in 100 µL TE buffer (10 mM Tris-HCl pH 8.0, 0.1 mM EDTA) and then preserved at -20°C until used.

ITS region of the rDNA was amplified using forward primer ITS1 (5'-TCCGTTGGTGAACCAGCGG-3') and reverse primer ITS4 (5'-TCCTCCGCTTATTGATATGC-3') (White, Bruns, Lee, & Taylor, 1990). The amplification reaction was carried out in 50 µL mixture containing respectively: 50 ng/µL of the template DNA, 2.5 µL of 0.2 mM of each primer, 0.5 µL of 2.5 U *Taq* DNA polymerase, 5 µL of 10x *taq* polymerase buffer, 0.1 µL of 100 Mm of each deoxy-nucleoside triphosphate (dNTPs), and 36.4 µL of deionised water. Thirty-four cycles were performed during PCR reaction with a starting denaturation step of 5 min at 94°C and a final extension step of 10 min at 72°C in a

thermocycler. The conditions of each cycle consisted of 30 s at 94°C, followed respectively by 30 s at 55°C and 1 min at 72°C for annealing and extension. Amplified products were separated on 1.2% agarose gel in TAE buffer, pre-stained with ethidium bromide (1 µg/mL) and electrophoresis was carried out at 60v for 3 h in TAE buffer. One Kb ladder was used as a marker. The gel was observed in a trans-illuminator over ultra violet light. The desired bands were cut from the gel with minimum quantity of gel portion using QIAGEN gel extraction kit. The purified DNA was sequenced using ITS1 and ITS4 primers.

BLAST program (www.ncbi.nlm.nih.gov/BLAST) were used to compare the 5.8S rRNA gene sequence of the test bio-agents against nucleotide collection database. Multiple sequence alignment and pairwise alignment were conducted between selected sequences from database which show at least 90% homology with sequences of antagonistic fungi. This alignment was performed using CLUSTAL W (Thompson, Higgins, & Gibson, 1994). Thereafter evolutionary distance matrixes between all the nucleotide sequences were generated. Phylogenetic evolutionary tree was then constructed by UPGMA (Unweighted Pair-group Method with Arithmetic mean) method using MEGA 7 (Molecular Evolutionary Genetics Analysis) program (Kumar, Stecher, & Tamura, 2016).

2.4. Evaluation of the effect of volatile and non-volatile compounds

The effect of volatile compounds produced by *Trichoderma* on pathogens was evaluated according to the method described by Dennis and Webster (1971a). A 5 mm mycelial disc of three days pre-culture of *Trichoderma* was inoculated on the centre of 90 mm PDA plate. The cover of the Petri dish was replaced with dish centrally inoculated with pathogen. Paraffin tap was used to seal together the two plates and the whole incubated at 28°C for 3 days. In control plate, the antagonist was replaced with an agar disc. The effect of non-volatile compounds was evaluated according to the method described by Dennis and Webster (1971b). In this case, each *Trichoderma* was grown for 48 h on a sterile cellophane disc laying on PDA in 90 mm Petri dish. The antagonist was replaced by an agar disc in the control plate. The cellophane with the mycelia was then removed and the test pathogen inoculated for an incubation period of 6 days at 28°C.

For both assay, each treatment consisted of four replications and the experiments was repeated twice. At the end of incubation, colony diameter of the pathogen was measured and the inhibition of mycelial growth evaluated using the last mentioned Equation (1).

2.5. Assessment of extracellular enzyme production

Hydrolytic activities of chitinase, cellulase, lipase and protease were detected on specific solid media. *Trichoderma* isolate was grown on a medium containing the enzyme substrate and the zone of degraded substrate (halos) formed around the colony was measured after 3 days of incubation. Chitinase activity was performed according to the method described by Agrawal and Kotasthane (2012). The detection medium was prepared using: 4.5 g/L of colloidal chitin, 3.0 g/L (NH₄)₂SO₄, 0.3 g/L MgSO₄, 2.0 g/L K₂HPO₄, 1 g/L citric acid monohydrate, 15 g/L agar, 0.15 g/L bromocresol purple, 200 µL of tween 80, pH 4.7 and autoclaved at 121°C for 15 min. Cellulase activity was determined by using plate screening medium containing 1% carboxymethylcellulose (Sigma, USA) (Florenco, Couri, & Farinas-Sanchez, 2012). The inoculated plates were incubated at 28°C and

thereafter stained with 0.1% Congo red dye solution for 15 min; the solution was discarded and the cultures washed with 1M NaCl for 15 min. Lipase activity was conducted according to Singh, Gupta, Goswami, and Gupta's (2006) method. *Trichoderma* were grown on chromogenic substrate plates containing for 1 L: phenol red (0.01%), 2% agar, 1% lipidic substrate (olive oil) and 10 mM CaCl₂ with the pH adjusted to 7.4 using 0.1 N NaOH. Protease activity was evaluated in agar plate containing 2.0 g/L K₂HPO₄, 10 g/L glucose, 5 g/L peptone, 15 g/L gelatin and 15 g/L agar. After incubation, Petri plates were floated with sodium sulphate (10%) (Berg et al., 2005). For all enzyme activities tests, the bioassays were done in 4 replicates and the experiment repeated twice.

2.6. In vivo biocontrol ability

Based on its relatively high hydrolytic enzyme production, only *T. gamsii* It-62 was selected for *in vivo* experiments. Common bean (*Phaseolus vulgaris* var GLP190C) seeds were surface-sterilised by using sodium hypochlorite (1%) and rinsed three times with distilled water. Forty-five grams (45 g) of seeds were soaked for 5 min in conidial suspensions (10⁶ conidia/mL) of *T. gamsii* It-62 containing 3% of carboxymethylcellulose (Merck, USA) (Khare, Singh, & Upadhyay, 2010). The seeds were then air dried in a laminar flow hood and those of seeds soaked only in carboxymethylcellulose were used as control.

For preparation of pathogen-inoculum, five mycelial disc (5 mm diameter) taken from one-week-old culture of a given pathogen were introduced into a 500 mL glass bottle containing 100 g of sterilised parboiled rice. After incubation at 28°C for 21 days, the content of the bottles were thoroughly mixed and used as different pathogens inoculum. Black peat substrate containing 160, 22.5 and 115 kg/ha of K, P and N with pH 6.01 was preliminary autoclaved for 2 h at 121°C and 15 psi. Thereafter, substrate was infested at the rate of 3% (v/v) with inoculum. Three hundred grams (300 g) of treated substrate were introduced into plastic pots (18 cm × 18.5 cm) and five treated seeds were sown in each pot and disposed in non-randomised design. Each treatment in triplicate consisted of four pots and the experiment was repeated twice. Ten different treatments namely T1 = *F. solani* + *T. gamsii* It-62; T2 = *F. oxysporum* + *T. gamsii* It-62; T3 = *M. phaseolina* + *T. gamsii* It-62; T4 = *P. ultimum* + *T. gamsii* It-62; T5 = *F. solani*; T6 = *F. oxysporum*; T7 = *M. phaseolina*; T8 = *P. ultimum*; T9 = *T. gamsii* It-62; and T10 = Control (untreated seeds and substrate) were arranged in nursery.

The disease incidence (%) was determined according to Abdel-Monaim, Abo-Elyousr, and Morsy (2011) and Saleh, Hoda, and Abd-El-Moneem (2016) in which pre-emergence and post-emergence damping-off was recorded 45 days after sowing. The relation between the number of non-emerged seeds and the number of sown seeds was considered as pre-emergence damping-off (2) whereas the relation between the number of plants showing disease symptoms and the number of emerged seedlings was considered as post-emergence damping-off (3).

$$\text{pre emergence damping-off} = \left(\frac{\text{non-emerged seedlings}}{\text{sown seeds}} \right) \times 100 \quad (2)$$

$$\text{post-emergence damping-off} = \left(\frac{\text{plants showing disease}}{\text{emerged seedlings}} \right) \times 100 \quad (3)$$

2.7. Statistical analyses

Data were analysed statistically by using STATGRAPHICS Computer Software, version 5.0. The differences among treatments for all the studied characteristics mentioned above were analysed for statistical significance using variance analysis (ANOVA). The means of treatment were compared by using Duncan's multiple range test at the $P < 0.05$ significance level.

3. Results

3.1. In vitro confrontation profiles of biocontrol agents vs. pathogens

From soil samples collected in bean rhizosphere, 85 *Trichoderma* spp. were isolated. All these antagonists significantly inhibited ($P < 0.05$) the radial growth of the pathogens. The inhibition rate varied from 54.4% to 79.4% depending on the pathogen and *Trichoderma* isolate. The highest level of antagonist (R1) with respect to the 4 pathogens tested was observed in proportions of 22.3%; 21.2%; 25.9% and 17.6%, respectively for *F. oxysporum*, *F. solani*, *M. phaseolina* and *P. ultimum* (Figure 1). Isolates It-13, It-21, It-58, It-62, P-8 and P-11 exhibited this class R1 antagonism against all the 4 pathogens were selected for further studies. Worthy of note, light microscopic observations made from the intermingled regions between the paired microorganisms on culture plate revealed the direct parasitism of *Trichoderma* on the pathogens. Any attempt to recover microorganisms from these regions led to the growth of *Trichoderma* only showing that the pathogens were killed.

3.2. Molecular identification and phylogenetic analysis

The six selected *Trichoderma* isolates were preliminarily characterised through morphological observations. Their genomic DNA was then amplified using ITS1 and ITS4 primers of the rDNA region. A fragment of about 600 bp in size was amplified with success from DNA of investigated bio-agents. The Blast search tool of GenBank available in NCBI allowed identifying fungi. In fact, the rDNA sequences of isolates were deposited in GenBank and accession number and percent similarity to closest strains were recorded. The percentage of similarity between antagonists and closest strains were in the range of 90% to 100%. The phylogenetic tree (Figure 2) generated by UPGMA analyses allowed to distribute strains into three different clades namely viride, hamatum and harzianum. The clade viride supported by 93% of bootstrap includes isolates It-21, It-58 and It-62 which were closely related to *T. koningiopsis* NBAII Ta-23, *T. erinaceum* NBAII TN-9 and *T. gamsii* XYX1-1, respectively. The clade hamatum supported by bootstrap values of up to 99% includes isolate It-13 closely related to his homologous *T. asperellum* SDLA27. The clade harzianum supported by bootstrap values of 100% includes isolates P-8 and P-11 which were closely related to *T. afroharzianum* CTCCSJ-G-HB40350 and *T. harzianum* VSL291, respectively.

3.3. Antibiosis profiles of the selected biological agents

The six selected isolates of *Trichoderma* corresponding to *T. asperellum*, *T. koningiopsis*, *T. erinaceum*, *T. gamsii*, *T. afroharzianum* and *T. harzianum* were tested for the

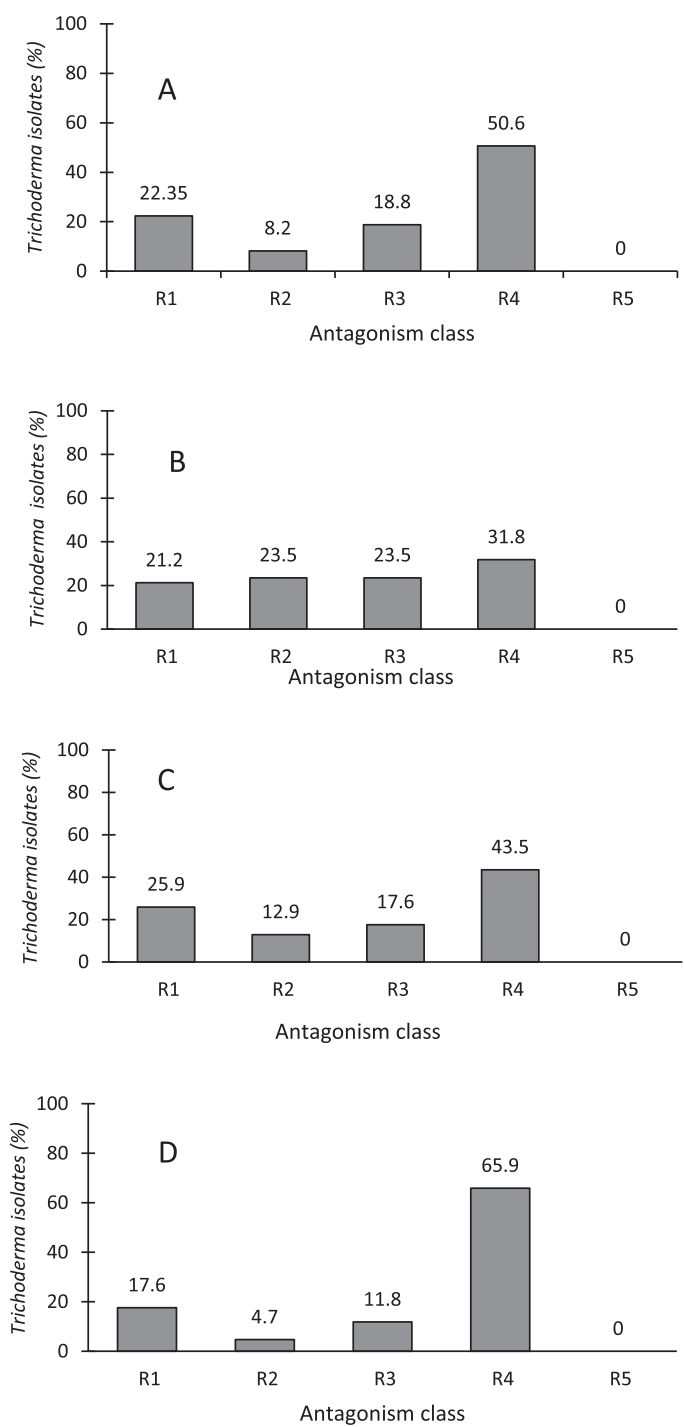


Figure 1. Proportion in percentage of different antagonist classes obtained during in-vitro dual culture with (A) *Fusarium oxysporum*; (B) *Fusarium solani*; (C) *Macrophomina phaseolina* and (D) *Pythium ultimum*.

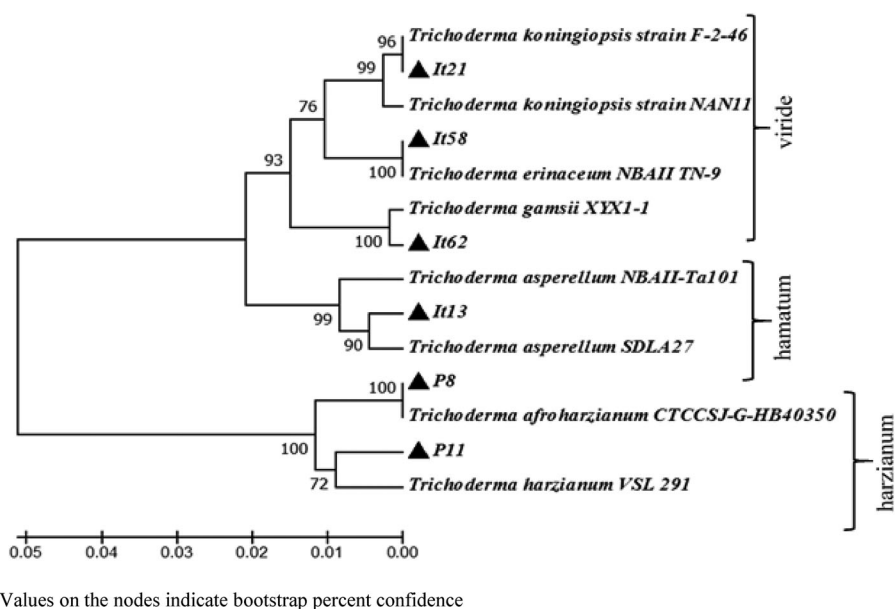


Figure 2. Phylogenetic analysis of *Trichoderma* isolates using MEGA7 with 1000 number of replications and UPGMA statistical method.

production of bioactive compounds. The results indicated that all these strains produced volatile and non-volatile molecules (Tables 1 and 2) that showed significant ($P < 0.05$) inhibitory effects against mycelial growth of the 4 common bean plant pathogens. The inhibition rate varied from 14.4% to 67.2% by volatile compounds and 13.7% to 75.9% by non-volatiles depending on the antagonists and the tested pathogens.

3.4. Production of cell wall degrading enzymes

Hydrolytic enzymes activities were characterised by halo zone formation around *Trichoderma* colony. The zone diameter ranged from 28.3 mm to 79.0 mm for chitinase, 50.0 mm to 63.8 mm for cellulase, 0.0 mm to 62.2 mm for protease and 36.3 mm to 60.7 mm for lipase (Table 3). *T. harzianum* P-11 exhibited the highest chitinase and

Table 1. Inhibitory effect of *Trichoderma* species on plant pathogens through production of volatile metabolites.

Species	Growth inhibition percentage (%)			
	Foxy	Fsol	Mpha	Pult
<i>T. asperellum</i> It-13	14.4 ± 1.1 ^e	17.5 ± 3.4 ^d	28.9 ± 1.9 ^e	51.1 ± 2.9 ^c
<i>T. koningiopsis</i> It-21	17.0 ± 1.7 ^d	21.4 ± 2.6 ^c	37.04 ± 1.7 ^d	44.1 ± 4.5 ^d
<i>T. erinaceum</i> It-58	29.3 ± 1.7 ^c	34.4 ± 1.1 ^b	52.2 ± 2.2 ^a	63.0 ± 5.5 ^b
<i>T. gamsii</i> It-62	33.7 ± 0.6 ^b	44.4 ± 1.1 ^a	48.1 ± 0.6 ^b	64.8 ± 1.7 ^b
<i>T. afroharzianum</i> P-8	41.8 ± 1.7 ^a	44.8 ± 0.6 ^a	45.4 ± 3.1 ^c	67.2 ± 1.5 ^a
<i>T. harzianum</i> P-11	34.8 ± 1.7 ^b	40.4 ± 5.6 ^a	43.3 ± 1.1 ^c	65.6 ± 1.1 ^b
CV (%)	37.3	35.6	19.2	15.4

Notes: Values are average of 4 replicates followed by standard deviation. Means within columns followed by the same letter (s) are not significantly different according to Duncan's multiple range test ($P < 0.05$). Foxy = *Fusarium oxysporum*, Fsol = *Fusarium solani*, Mpha = *Macrophomina phaseolina*, and Pult = *Pythium ultimum*. CV: Coefficient of variation.

Table 2. Inhibitory effect of *Trichoderma* species on plant pathogens through production of non-volatile metabolites.

Species	Growth inhibition percentage (%)			
	<i>Foxy</i>	<i>Fsol</i>	<i>Mpha</i>	<i>Pult</i>
<i>T. asperellum</i> It-13	13.7 ± 1.7 ^d	20.4 ± 3.9 ^e	34.1 ± 5 ^c	21.1 ± 3.8 ^e
<i>T. koningiopsis</i> It-21	26.3 ± 2 ^c	27.7 ± 1.1 ^d	27.0 ± 2.8 ^e	24.5 ± 5.5 ^e
<i>T. erinaceum</i> It-58	22.9 ± 3.9 ^c	32.5 ± 2.3 ^c	35.2 ± 6.3 ^c	29.6 ± 1.7 ^d
<i>T. gamsii</i> It-62	31.8 ± 7.1 ^b	30.0 ± 1.1 ^c	31.1 ± 1.9 ^d	38.9 ± 2.2 ^c
<i>T. afroharzianum</i> P-8	63.7 ± 8 ^a	57.0 ± 0.6 ^b	68.1 ± 1.7 ^b	65.6 ± 1.1 ^b
<i>T. harzianum</i> P-11	65.9 ± 0.6 ^a	69.6 ± 1.7 ^a	75.9 ± 2.6 ^a	69.3 ± 2.6 ^a
CV (%)	59	48.9	46.4	50.6

Notes: Values are average of 4 replicates followed by standard deviation. Means within columns followed by the same letter (s) are not significantly different according to Duncan's multiple range test ($P < 0.05$). *Foxy* = *Fusarium oxysporum*, *Fsol* = *Fusarium solani*, *Mpha* = *Macrophomina phaseolina*, and *Pult* = *Pythium ultimum*. CV: Coefficient of variation.

Table 3. Holo-enzyme activities exhibited by selected strains of *Trichoderma* on solid media after 3 days incubation at 28°C.

Species	Chitinase	Holo-enzyme zone (mm)		
		Cellulase	Protease	Lipase
<i>T. asperellum</i> It-13	58.7 ± 2.3 ^b	53.8 ± 0.8 ^c	32.0 ± 2.6 ^a	50.7 ± 2.1 ^b
<i>T. koningiopsis</i> It-21	51.3 ± 3.2 ^{ab}	50.3 ± 0.6 ^{ab}	24.0 ± 2.0 ^{ab}	36.3 ± 1.5 ^{ab}
<i>T. erinaceum</i> It-58	28.3 ± 3.8 ^c	50.0 ± 2.7 ^{ab}	12.0 ± 1.7 ^c	38.7 ± 1.5 ^{ab}
<i>T. gamsii</i> It-62	57.3 ± 1.1 ^b	51.7 ± 1.5 ^{ab}	60.2 ± 1.3 ^b	50.0 ± 2.6 ^b
<i>T. afroharzianum</i> P-8	73.3 ± 0.6 ^a	61.8 ± 0.3 ^b	13.0 ± 1.0 ^{ab}	60.7 ± 1.2 ^a
<i>T. harzianum</i> P-11	79.0 ± 3.0 ^a	63.8 ± 0.8 ^a	0.0 ± 0.0 ^{bc}	59.0 ± 1.0 ^a
CV (%)	30.9	10.9	89.4	20.3

Notes: Values are average of 4 replicates followed by standard deviation. Means within columns followed by the same letter are not significantly different according to Duncan's multiple range test ($P < 0.05$). CV: Coefficient of variation.

cellulase activity while the greatest protease and lipase activity was recorded with *T. gamsii* It-62 and *T. afroharzianum* P-8, respectively. Therefore, the highest mean score obtained by calculating the mean of the 4 hydrolytic enzymes production was recorded with *T. gamsii* It-62 (54.8 mm).

3.5. Ability of *T. gamsii* to control damping-off disease

Data presented in Table 4 showed that there was a significant ($P < 0.05$) decrease in damping-off disease incidence when common bean seeds were treated by *T. gamsii* It-62 as compared to uninoculated seedlings. The percentage of disease control (Figure 3) exhibited by *T. gamsii* were 63.6%; 75%; 80% and 67.9% at pre-emergence and 74.3%; 80.9%; 68.2% and 87.5% at post-emergence in the presence of *F. oxysporum*, *F. solani*, *M. phaseolina* and *P. ultimum* respectively.

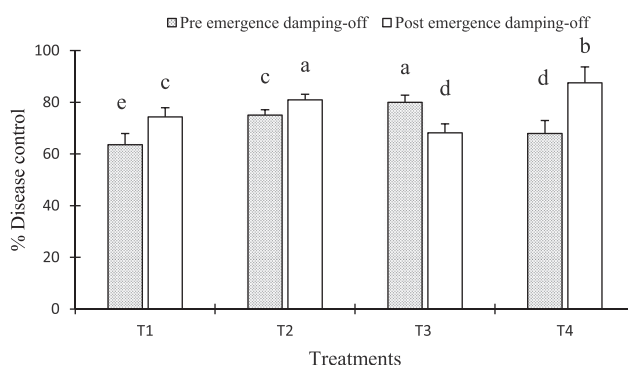
4. Discussion

Soil borne fungal pathogens causing damping-off disease are a serious constraint to common bean production as they affect seedling establishment leading to weak emergence and delayed development of the plant. Intensive research on the management of common bean damping-off has focused on diverse strategies including biological control. In this study, 85 *Trichoderma* isolates obtained from common bean rhizospheric soil were first

Table 4. Effect of *Trichoderma gamsii* It-62 on incidence of pre and post-emergence damping-off of common bean 45 days after sowing.

Treatment	%Damping-off	
	Pre-emergence	Post-emergence
(T1) <i>F. oxysporum</i> + <i>T. gamsii</i>	20 ± 2.3 ^{cd}	20 ± 3.2 ^{cd}
(T2) <i>F. solani</i> + <i>T. gamsii</i>	15 ± 3 ^e	16.7 ± 4.8 ^{cd}
(T3) <i>M. phaseolina</i> + <i>T. gamsii</i>	7 ± 2.5 ^{de}	17.6 ± 3 ^{cd}
(T4) <i>P. ultimum</i> + <i>T. gamsii</i>	25 ± 5 ^d	12.5 ± 2.5 ^e
(T5) <i>F. oxysporum</i>	55 ± 3.1 ^{ab}	77.8 ± 3 ^{ab}
(T6) <i>F. solani</i>	60 ± 4.8 ^b	87.5 ± 4.2 ^b
(T7) <i>M. phaseolina</i>	35 ± 2.5 ^c	55.3 ± 5.3 ^c
(T8) <i>P. ultimum</i>	78 ± 3 ^a	100 ± 0 ^a
CV (%)	25	33.8

Notes: Values are average of 4 replicates followed by standard deviation. Means within columns followed by the same letter (s) are not significantly different according to Duncan's Multiple Range ($P < 0.05$). CV= coefficient of variation.

**Figure 3.** Efficacy of *Trichoderma gamsii* It-62 on common bean damping-off caused by *Fusarium solani*, *Fusarium oxysporum*, *Macrophomina phaseolina*, and *Pythium ultimum* 45 days after sowing.

screened *in vitro* by dual culture test to identify effective bio-agents against *F. solani*, *F. oxysporum*, *M. phaseolina* and *P. ultimum*. All these antagonists significantly inhibited the radial growth of the pathogens and six of them were able to overgrow completely the mycelia of plant pathogens. They were successfully identified by molecular approach using ITS regions of ribosomal genes for the construction of primers. ITS region of rDNA was amplified using specific ITS-1 and ITS4 primers. Amplified products of 600 bp size were produced by the primers. This result is in agreement with several workers which obtained successfully a fragment of 600 pb after amplification of the 5.8-rDNA region in *Trichoderma* (Castrillo, Bich, Zapata, & Villalba, 2016; Chakraborty, Chakraborty, Saha, Dey, & Sunar, 2010). The results obtained from BLAST search tools allowed us to identify the different isolates to species level with homology percentage of at least 99%. The results obtained from phylogenetic tree are in accordance with previous investigations concerning the topology of *Trichoderma* phylogeny (Hermosa et al., 2000).

In dual cultures, an attempt to recover the pathogens from the intermingling regions failed, showing that they were killed. The rapid growth and spatial colonisation of *Trichoderma* observed in may be possible mechanism of biological control. In fact, *Trichoderma* spp. are known as fast-growing fungi. This rapid growth gives them an important advantage in the competition for space and nutrients with plant pathogenic fungi (Barbosa,

Rehn, Menezes, & Mariano, 2001; Sid Ahmed, Perez-Sanchez, Egea, & Candela, 1999). Besides this mechanism, several key factors which chiefly contribute to antagonism of these fungi were already reported and these include mycoparasitism, antibiosis by lytic enzymes and secondary metabolites secretion (Harman, 2006; Verma, Brar, Tyagi, Surampalli, & Valero, 2007; Vinale et al., 2007). In the present study, it was also found an evidence of production of inhibitory compounds and enzymes as well as direct coils of *Trichoderma* around some pathogens. Similar findings have been reported by Khaledi and Taheri (2016) who studied the antagonism of *T. harzianum* against *M. phaseolina* and suggested the production of volatile compounds as one of the main mechanism. In the same line, Cherkupally, Amballa, and Reddy (2017) and Meena, Swapnil, Zehra, and Dubey (2017) also reported the involvement of volatile and non-volatile compounds produced by *T. harzianum* and *T. viride* against *Fusarium* spp. and *Alternaria* spp. Cell wall degrading enzymes are known to be responsible for the destruction of cell integrity of pathogens by *Trichoderma* (Kubicek et al., 2001). However, according to Tahía, Rincón, and Carmen (2004), bio-control results from the combination of different mechanisms which includes physical contact, competition for nutrients and space, the production of lytic enzymes and antibiotics. These mechanisms are used sequentially or/and simultaneously (Vinale et al., 2008). Despite the fact that, it is very difficult to predict the behaviour in natural environments of a given strain having laboratory bio-control activity, *in vitro* assays are useful to determine antagonistic profiles and select potential bio-control agents against particular plant pathogens before assessing glasshouse and field trials (Hermosa et al., 2000).

Trichoderma spp. are well known as biological control agents because of their ability to reduce the population of a wide range of soil-borne plant pathogens (Heidi, 2012; Nargund, Amaresh, Sreenivas, & Nadagouda, 2007). In this study, seeds treatment with *T. gamsii* showed a beneficial effect on the emergence of bean plants by inhibiting damping-off incidence and delaying the onset of disease symptoms induced by the different plant pathogens. These results are consistent with those obtained by other authors who shown that fungi of the genus *Trichoderma* are able to reduce the incidence of damping-off caused by pathogenic fungi on bean seedlings (Abd-El-Khair, Khalifa, & Hagga, 2010; Malik, Dawar, Sattar, & Dawar, 2005; Ruvimbo, Chaibva, Takawira, & Njeni, 2016). Abd-El-Khair et al. (2010) reported that seeds treated with *T. album*, *T. hamatum*, *T. harzianum* and *T. viride* were effective in controlling damping-off in common bean caused by *F. solani* and *R. solani*. Similarly, Toghueo et al. (2016) showed that *T. atroviridae* reduced disease incidence on bean seedlings caused by *F. solani*. However, this work is the first report referring to the antagonistic effect of *T. gamsii* on soil-borne pathogens complex (*F. oxysporum*, *F. solani*, *M. phaseolina* and *P. ultimum*) causing damping-off on common bean in Cameroon. Several modes of action have been proposed to explain the biological control potential of plant pathogenic agents by *Trichoderma* spp *in vivo*. These include particularly direct antagonism by the competition for space and nutrient, secretion of cell wall degrading enzymes and antibiosis. Besides these mechanisms, activation of the plant's defense mechanisms by inducing metabolite pathways responsible for the expression of localised or systemic resistance to disease in plant is also well documented (Harman et al., 2004; Mayo et al., 2015; Rao, Reddy, & Surekha, 2015; Singh, Singh, Singh, & Singh, 2013; Yedidia, Benhamou, & Chet, 1999). In this line, *Trichoderma* produces a wide spectrum of secondary metabolites

that are known to serve as triggers of systemic resistance against a broad range of plant pathogens (Rao et al., 2015). Root colonisation by different *Trichoderma* spp. like *T. asperellum*, *T. hamatum*, *T. harzianum* and *T. virens* have triggered systemic acquired resistance (SAR) against different diseases (Harman et al., 2004). Also, *Trichoderma* is able to colonise plant root tissues and during root colonisation, the biological agent develops in the intercellular cortex and plant root vessels without altering their structures. Once in these, it is then not only able to come into contact with the pathogen during an attack but also can induce localised or systematic resistance to diseases through the release of metabolites (Harman et al., 2004; Vinale et al., 2008). The release of those chemicals tends to promote plant production of ethylene or terpenoid phytoalexins, which are involved in plant resistance to pathogens (Howell, Hanson, Stipanovic, & Puckhaber, 2000). However, this work has to be exploring in depth to study the potential mechanisms that could explain and confirm the results obtained.

5. Conclusion

The screening of 85 *Trichoderma* isolates from the rhizospheric soil of common bean led to identified six (*T. asperellum* It-13, *T. koningiopsis* It-21, *T. erinaceum* It-58, *T. gamsii* It-62, *T. afroharzianum* P-8 and *T. harzianum* P-11) species with high antagonistic potential against *F. oxysporum*, *F. solani*, *P. ultimum* and *M. phaseolina*. These *Trichoderma* isolates produce volatile and non-volatiles compounds as well as hydrolytic enzymes. *T. gamsii* It-62 could be an excellent candidate for providing long-term biocontrol potential for the management of damping-off in common bean plant. Further field experiments will validate this candidate biocontrol agent as an alternative to chemical pesticide.

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